- ENGINEERING MATERIALS -

Dr. Nitesh Chindarkar

MODULE VI : Nanomaterials

(No. of hours – 04, Weightage of marks with option – 13)

- **Course Objective**: To study the types, properties and uses of various Nanomaterials.
- **Course Outcome**: Find different nanomaterials for the scientific applications.

MODULE VI : Nanomaterials

Sr. No.	TOPIC	REMARKS
1.	A) Definition, Types of Nanostructured materials, Applications of Nanomaterials.	Definition of Nanomaterial, Types of nanostructured materials **(Ref.1)** – (1) inorganic-based nanomaterials; (2) carbon-based nanomaterials; (3) organic-based nanomaterials; and (4) composite-based nanomaterials (1-2 Examples of each type). Applications of Nanomaterials.
2.	B) Graphene	Structure and properties.
3.	C) Types of Carbon Nanotubes (SWCNTs and MWCNTs) – Properties and Uses.	Explain types with their properties and uses.

Reference Book (Nanomaterials):

1. M. Nasrollahzadeh, S. M. Sajadi, M. Sajjadi, Z. Issaabadi, M. Atarod, *Interface Science and Technology*, Elsevier, Volume 28, 2019, Pages 29-80, ISSN 1573-4285, ISBN 9780128135860, https://doi.org/10.1016/B978-0-12-813586-0.00002-X (**for Types of Nanostructured materials**).

2. Engineering Chemistry. Jain & Jain, Dhanpat Rai Publishing Company

3. A Text book of Engineering Chemistry, Shashi Chawla, Dhanpat Rai & CO.

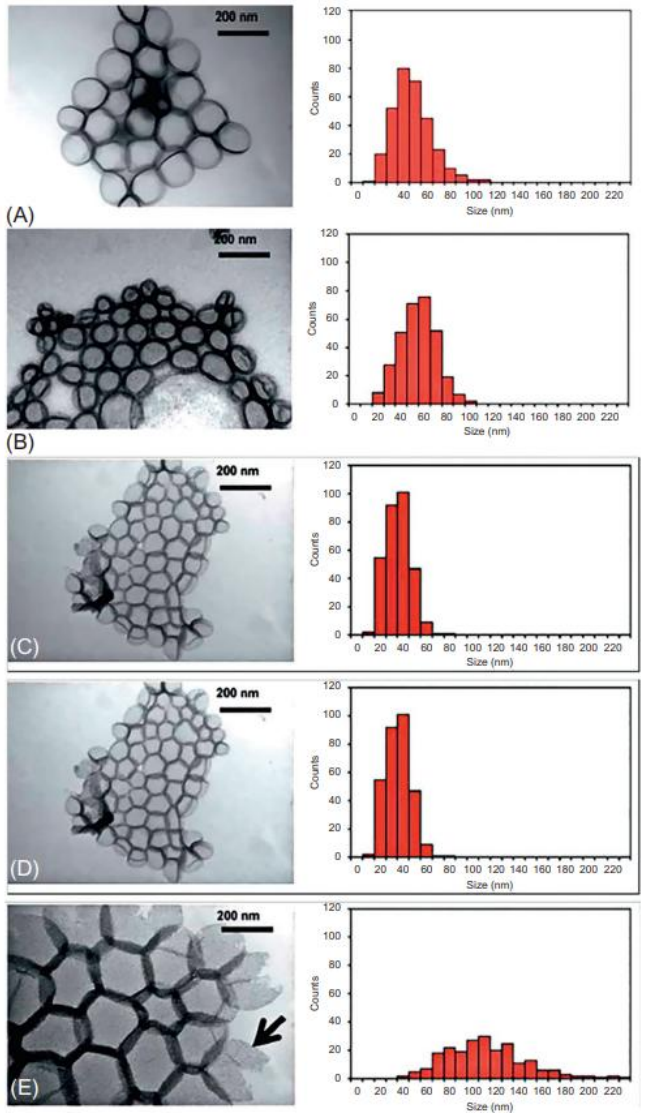
Nanomaterials

- Nanomaterials are materials with at least one external dimension in the size range of approximately **1-100 nanometers (nm)**.
- These materials exhibit unique physical, chemical, and biological properties due to their nanoscale dimensions, which differ significantly from their bulk counterparts.

Types of Nanostructured materials

(1) Inorganic-based nanomaterials:

- Inorganic nanostructures and nanoparticles are structures and particles that are not made from carbon-based or organic-based NPs.
- Metal-based and metal oxide-based NPs are commonly categorized as inorganic nanostructures.
- These nanostructures can be synthesized into **metal NPs, such as Pd or Au**, **metal oxide NPs like TiO_2, and also semiconductors, such as ceramics and silicon**.



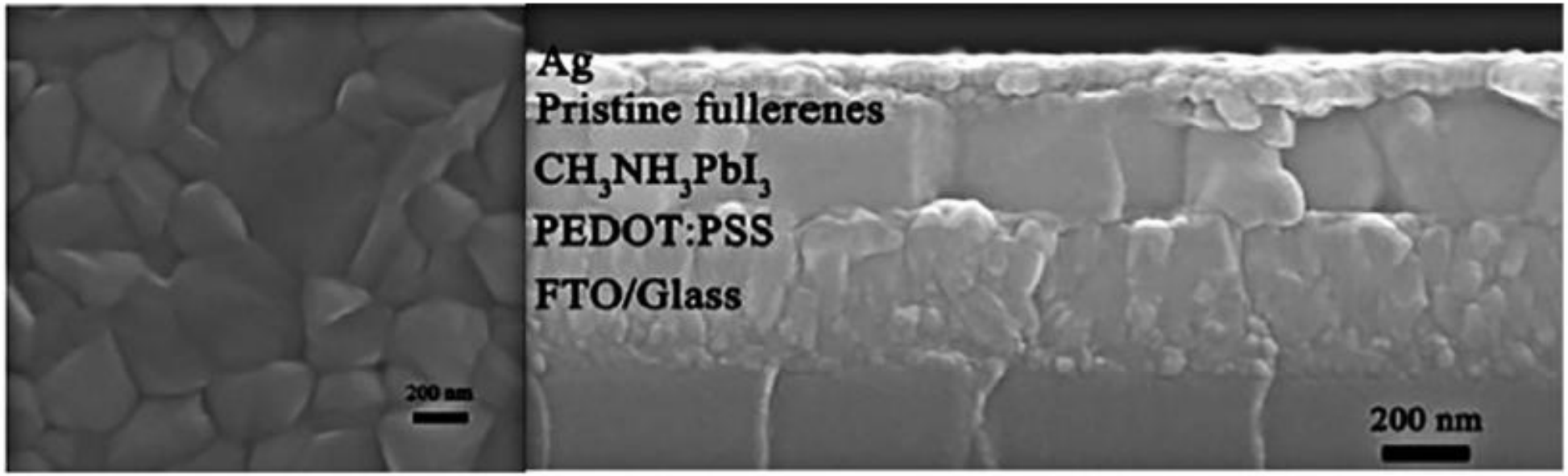
Inorganic-based nanomaterials (Silica Nanocapsules)

Types of Nanostructured materials

(2) Carbon-based nanomaterials:

- Nanostructures made of carbon are known as carbon-based nanostructures.
- They can have different morphologies, such as ellipsoid, hollow tube, or sphere.
- Generally, these nanostructures can be classified into diamonds, fullerenes (C_{60}, C_{80}, and C_{240}), carbon nanotubes (CNTs), graphene, and carbon nanofibers.

Carbon-based nanomaterials

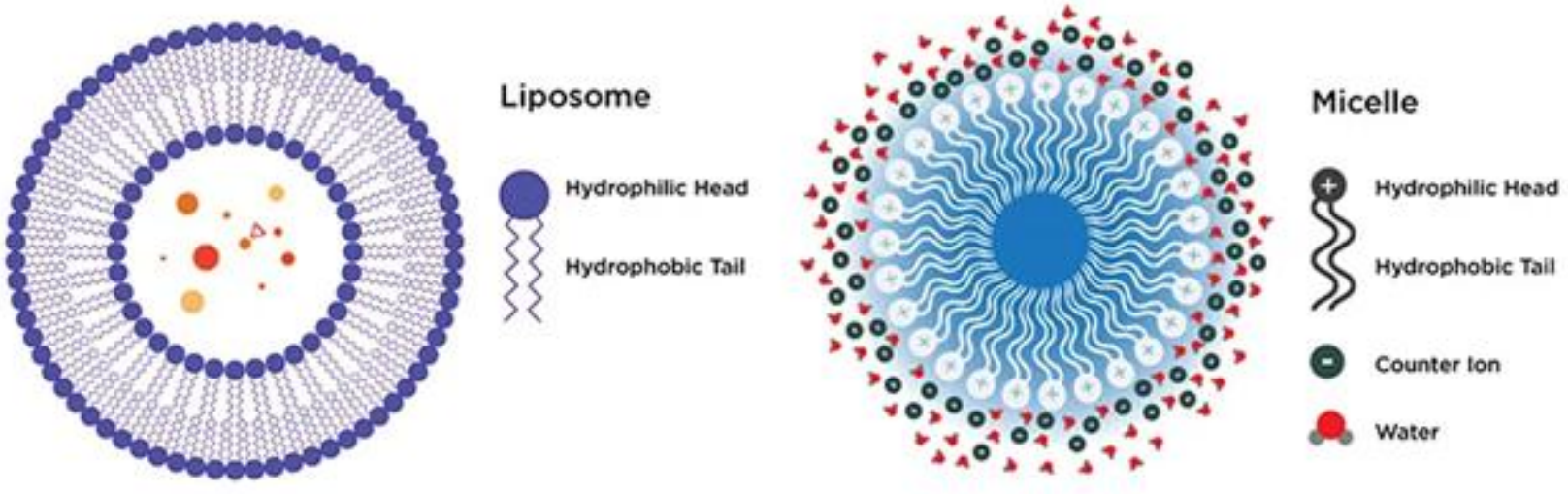


Types of Nanostructured materials

(3) Organic-based nanomaterials:

- Dendrimers, liposomes, micelles, polymer NPs, etc., are usually known as organic nanostructures or polymers.
- These include nanostructures made mostly from organic material, excluding carbon-based or inorganic-based nanostructures.
- These nanostructures are nontoxic, biodegradable, and some of them structures, e.g., liposomes and micelles, have hollow cores (also known as nanocapsules).

Organic-based nanomaterials

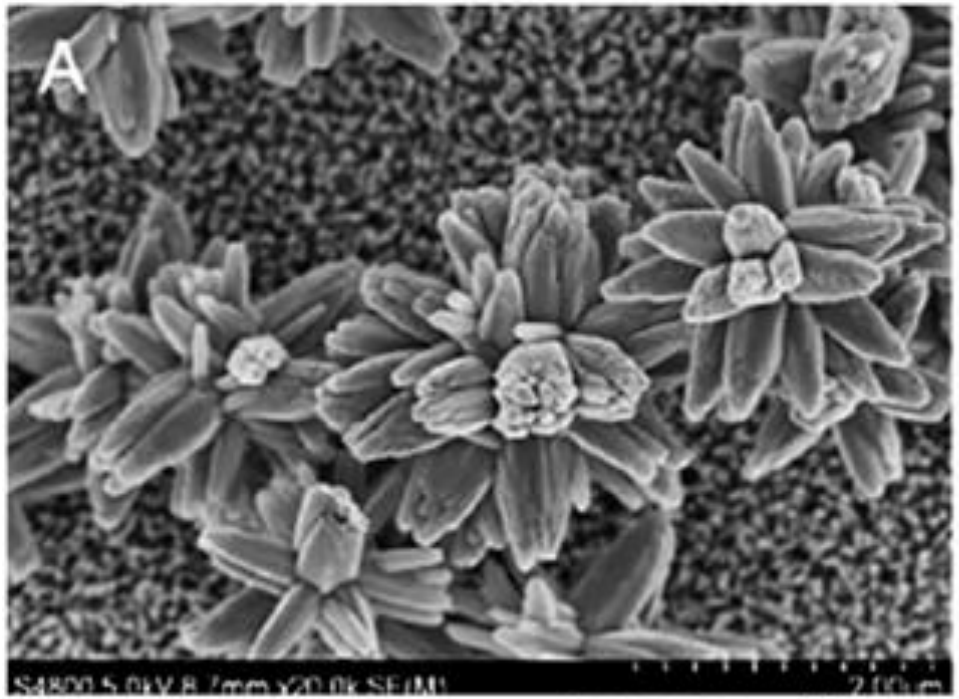


Types of Nanostructured materials

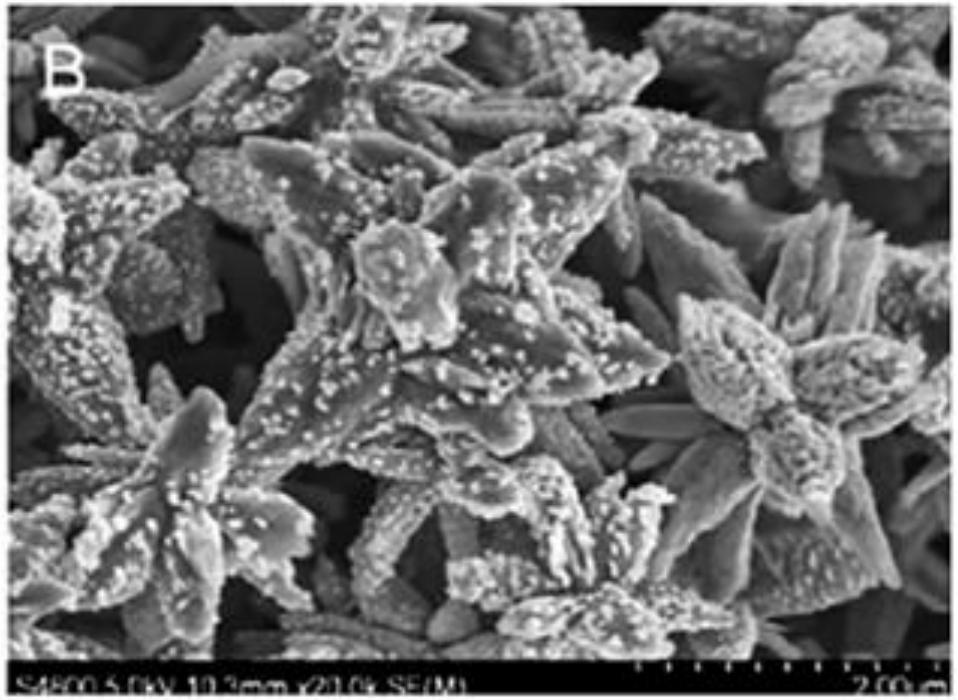
(4) Composite-based nanomaterials:

1. Nanocomposites can be described as multiphase nanostructures with one phase being at the nanoscale dimension.
2. They can either combine nanostructures with other NPs or NPs with bulk-type or larger materials (e.g., hybrid nonporous materials) or more complicated structures.
3. Nanocomposites can be any combination of metal-based, carbon-based, or organic-based nanostructures with any form of ceramic, metal, or polymer bulk materials.
4. Au-ZnO flower-rods (FRs), ZnO FRs.

Composite-based nanomaterials



(A) SEM image of ZnO FRs.



(B) SEM image of Au-ZnO FRs

Applications of Nanomaterials

- Nanomaterials have a wide range of applications across various fields, including:

1. **Electronics**: Nanoscale transistors, memory devices, and displays.
2. **Medicine**: Targeted drug delivery, cancer treatment, imaging, and diagnostics.
3. **Energy**: Solar cells, fuel cells, batteries, and supercapacitors.
4. **Environmental Remediation**: Water purification, air filtration, and pollution cleanup.
5. **Cosmetics**: Skincare products, sunscreens, and hair care.
6. **Textiles**: Waterproof, stain-resistant, and self-cleaning fabrics.

Applications of Nanomaterials

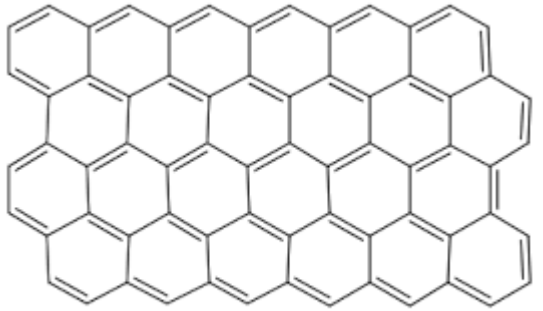
7. **Food Packaging**: Improved shelf life, antimicrobial properties, and sensing.

8. **Aerospace**: Lightweight composites, coatings, and propulsion systems.

9. **Biotechnology**: Biosensors, biochips, and tissue engineering.

10. **Catalysis**: Improved catalysts for chemical reactions and fuel production.

Graphene



Graphene

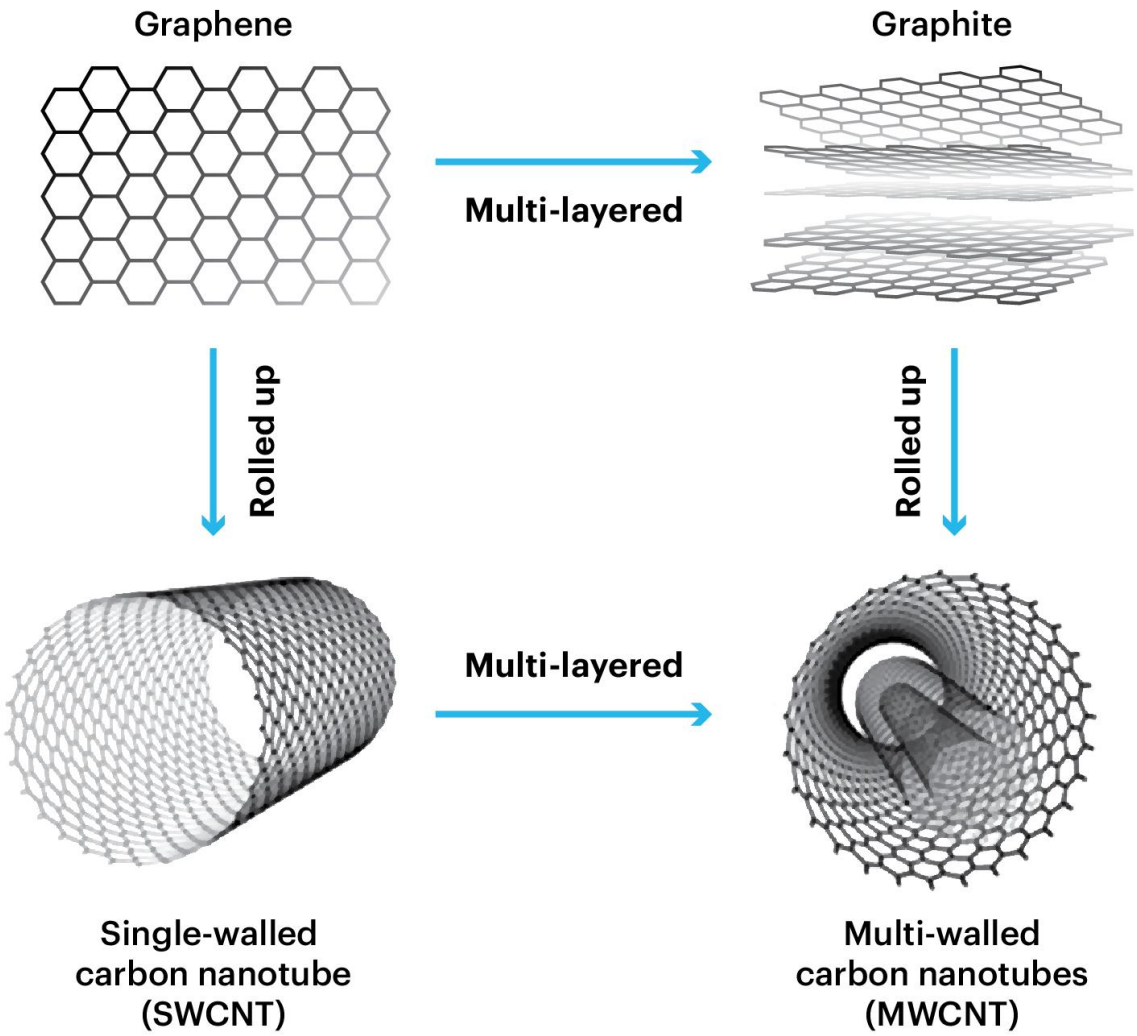
Structure:

1. **2D Material**: Graphene is a two-dimensional (2D) material composed of a single layer of carbon atoms.
2. **Hexagonal Lattice**: The carbon atoms are arranged in a hexagonal lattice structure, with each atom bonded to three neighboring atoms.
3. **sp2 Hybridization**: The carbon atoms in graphene exhibit sp2 hybridization, resulting in a planar, trigonal geometry.

Properties of Graphene

1. **High Electrical Conductivity**: Graphene is an excellent conductor of electricity, with a high carrier mobility.
2. **High Thermal Conductivity**: Graphene has a high thermal conductivity, making it suitable for heat management applications.
3. **Mechanical Strength**: Graphene is incredibly strong, with a Young's modulus of approximately 1 TPa.
4. **Optical Transparency**: Graphene is highly transparent, with an absorbance of approximately 2.3%.
5. **Chemical Stability**: Graphene is highly resistant to chemical corrosion and oxidation.
6. **High Surface Area**: Graphene has a high surface area, making it suitable for applications such as energy storage and sensing.

Types of Carbon Nanotubes (CNTs)



Types of Carbon Nanotubes (CNTs)

1. **Single-Walled Carbon Nanotubes (SWCNTs)**

- **Structure**: A single layer of carbon atoms rolled into a cylindrical shape.
- **Properties**:

- High electrical conductivity
- High thermal conductivity
- High mechanical strength
- High surface area

- **Uses**: Electronics (transistors, interconnects), Energy storage (batteries, supercapacitors), Biomedical applications (drug delivery, imaging)

Types of Carbon Nanotubes (CNTs)

2. Multi-Walled Carbon Nanotubes (MWCNTs)

Structure: Multiple layers of carbon atoms rolled into a cylindrical shape.

- **Properties**:

- High electrical conductivity
- High thermal conductivity
- High mechanical strength
- Lower surface area compared to SWCNTs

- **Uses**: Composites (reinforced polymers, ceramics), Energy storage (batteries, supercapacitors), Biomedical applications (tissue engineering, biosensors)


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